

Group-5:

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SDLC Model Selection for Sustainable Farm Decision Support and Crop Advisory System

1. Objective(s)

Recommend suitable crops based on soil, season, and cropping history. Schedule irrigation and suggest fertilizers to optimize resource use.

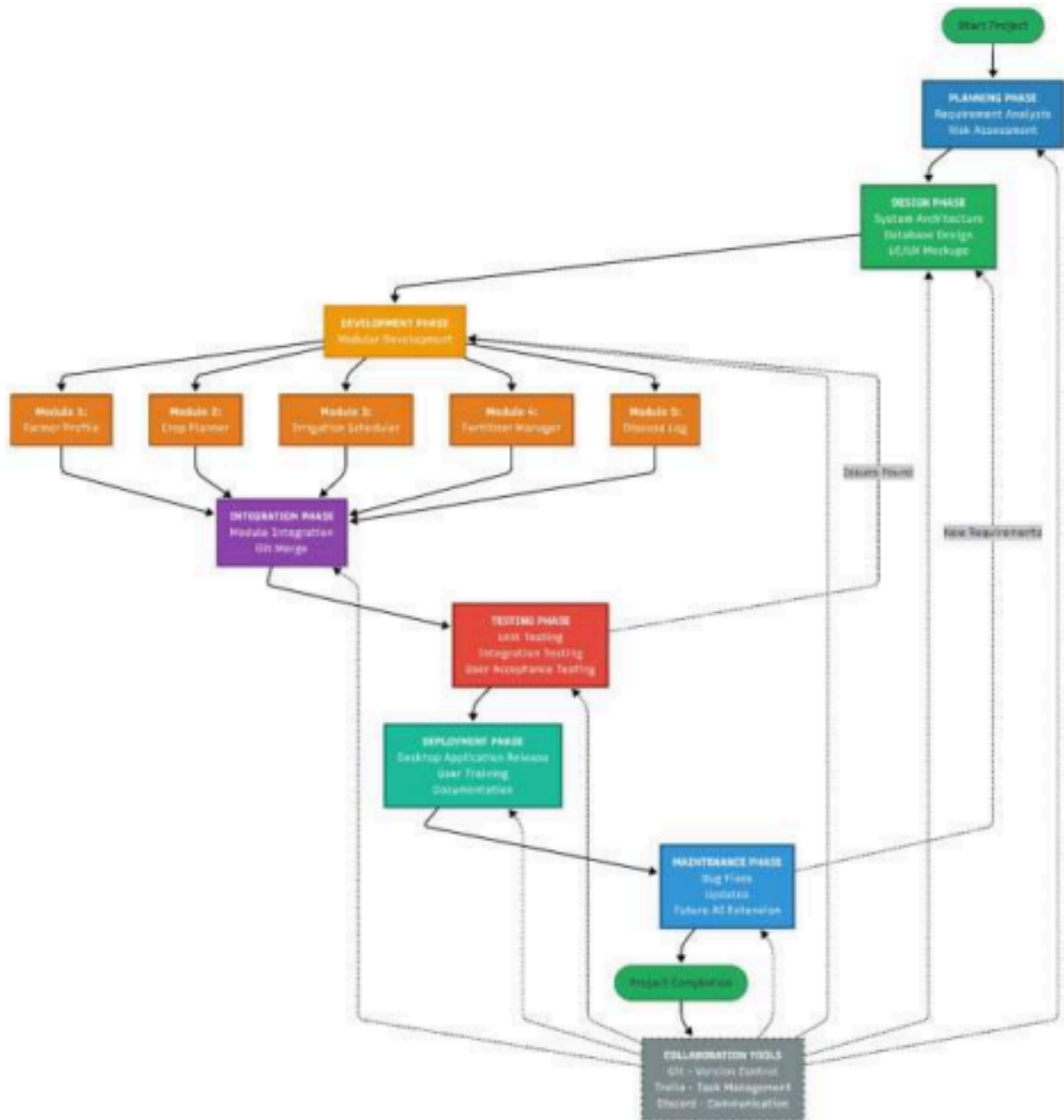
Log crop diseases and generate health reports for timely action.

Provide an intuitive interface for farmers with basic computer skills.

2. Problem analysis

The Sustainable Farm Decision Support and Crop Advisory System is designed to assist farmers in making informed decisions regarding crop selection, irrigation scheduling, fertilizer management, and crop health monitoring. Farmers often lack access to timely, personalized advice, leading to suboptimal resource usage and increased crop loss. The system will provide a digital platform where farmers can input farm details (land size, soil type, previous crops) and receive recommendations. It will also maintain logs for irrigation, fertilizer applications, and disease observations. Future enhancements include AI-based disease detection and yield prediction.

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3. Methodology

Step 1: Analyze the Problem

- System Purpose: Provide farmers with crop planning, irrigation scheduling, fertilizer management, and disease logging.
- Scope: Desktop application with modular architecture; future AI extensions planned.
- User Needs: Simple interface, reliable recommendations, maintainable records, collaborative development workflow.

Step 2: Evaluate Project Characteristics

- Requirement Stability: Core requirements are clear, but AI-based features (disease detection, yield prediction) will evolve later.

- Risk Level: Moderate – risks include incorrect user input, integration delays, and scalability.
- Complexity: Low to medium – rule-based logic initially; modular design simplifies development.
- Delivery Timeline: 8 weeks (4 sprints of 2 weeks each), with a working prototype expected early.
- Team & Collaboration: Small team using Git, Trello, Discord; frequent feedback needed from farmers.

Step 3: Compare SDLC Models

A comparison matrix was created using weighted criteria (priority 1–6). The table below summarizes the results:

Criteria	Waterfall	Incremental	Iterative	Spiral	Agile	Prototype
Priority	1	2	3	4	5	6
Requirement clarity (core fixed, AI evolves)	No	No	No	Yes	Yes	No
Modular development	No	No	Yes	Yes	Yes	Yes
Risk management	No	No	No	Yes	Yes	No
Team collaboration (Git/Trello)	No	No	Yes	No	Yes	No
Working prototype	No	No	Yes	Yes	Yes	Yes
Documentation overhead	Yes	Yes	No	No	No	No

3 Scalability

Criteria	Priority	Total	for future AI			
			V-Model Iterative	Spiral Agile	No Yes	No Yes Yes Yes
Waterfall						

Score 0 0 38 32 49 43

Step 4: Select Suitable Model

The Agile model achieved the highest total score (49). It supports evolving requirements, user feedback, modular delivery, and team collaboration. The Spiral model (32) is heavier and more suited to high-risk projects; Prototype (43) lacks structured iteration.

Step 5: Justify Selection

Agile is optimal because:

- Flexibility: Accommodates changes from farmer feedback and future AI modules.
- User-Centric: Farmers can test early increments, ensuring usability.
- Modular Delivery: Independent modules (crop planner, irrigation, etc.) can be developed, tested, and integrated incrementally.
- Team Fit: The small team's use of Git, Trello, and Discord aligns with Agile practices (sprints, daily stand-ups, task boards).
- Early Value: A working prototype (e.g., crop planner) can be delivered quickly, validating the concept before investing in full development.

This methodology ensures a structured yet adaptive approach to building the farm advisory system.